

AI Attendance System Using Face Recognition

DESIGN DOCUMENT

Overview of Design

The design phase describes how the AI Attendance System using Face Recognition is structured and implemented. The system is designed using modular architecture so that each component performs a specific function. The main goal of the design is to ensure accuracy, fast processing, ease of use, and scalability. The design includes architectural design, detailed design, UML diagrams, DFD models, and database structure.

The system uses computer vision and machine learning techniques to detect and recognize faces. Python is used as the programming language because it provides strong support for artificial intelligence and image processing libraries. OpenCV is used for face detection and image processing, while the face recognition library is used for identifying individuals based on facial features.

Design decisions are made based on performance, ease of implementation, cost, and accuracy. Compared to traditional biometric methods such as fingerprint and RFID cards, face recognition provides a contactless and secure solution. The trade-off study shows that face recognition provides better user convenience, prevents proxy attendance, and reduces hardware dependency.

Trade-off Analysis Summary:

Method	Advantages	Limitations
Manual Attendance	Simple	Time consuming, errors
RFID Cards	Fast	Can be misused
Fingerprint	Accurate	Contact required
Face Recognition	Contactless, secure	Needs camera

Based on the comparison, face recognition is selected as the best solution.

a. System Design

Architectural Design

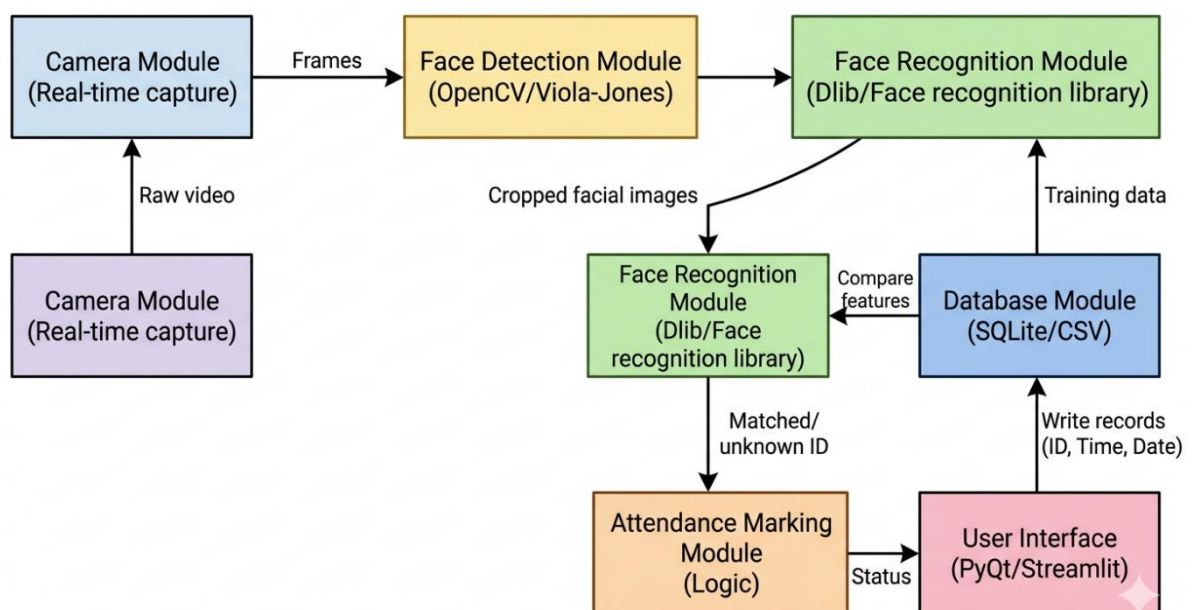
The system follows a modular architecture where each module performs a specific function. The main components of the system include image capture, face detection, face recognition, database storage, and attendance marking.

Architecture Components:

1. Camera Module – captures real-time images.
2. Face Detection Module – detects faces in the captured image.
3. Face Recognition Module – compares captured face with stored dataset.
4. Attendance Module – records attendance with date and time.
5. Database Module – stores user details and attendance records.
6. User Interface – allows interaction with the system.

Advantages of modular design:

- Easy to maintain
- Easy to upgrade
- Improves performance
- Reduces complexity



Data Flow Diagram (DFD)

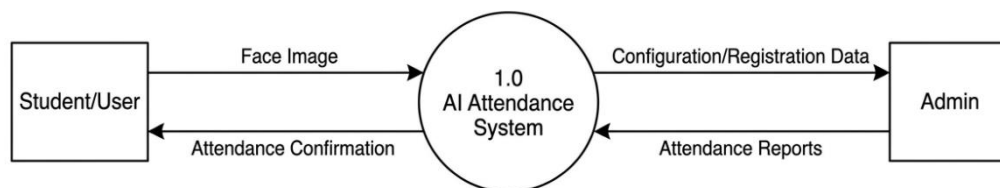
DFD shows how data moves through the system. The main data flow starts from capturing an image, processing the image, recognizing the face, and storing attendance data.

DFD Levels:

Level 0 – Overall system process

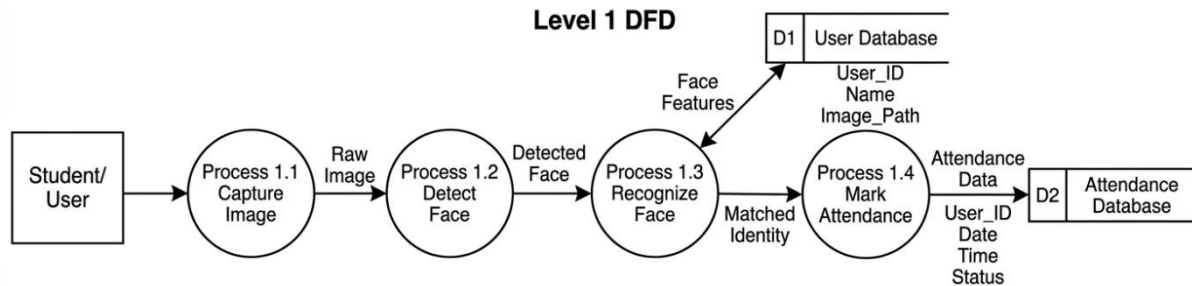
Level 1 – Detailed process of face recognition and attendance storage

Level 0 Context Diagram



Level 0 Context Diagram

Level 1 DFD



Level 1 DFD

UML Design

UML diagrams help in understanding system structure and interaction between components.

Use Case Diagram

Shows interaction between user and system.

Actors:

- Student/User
- Admin

Functions:

- Register face
- Train dataset
- Mark attendance

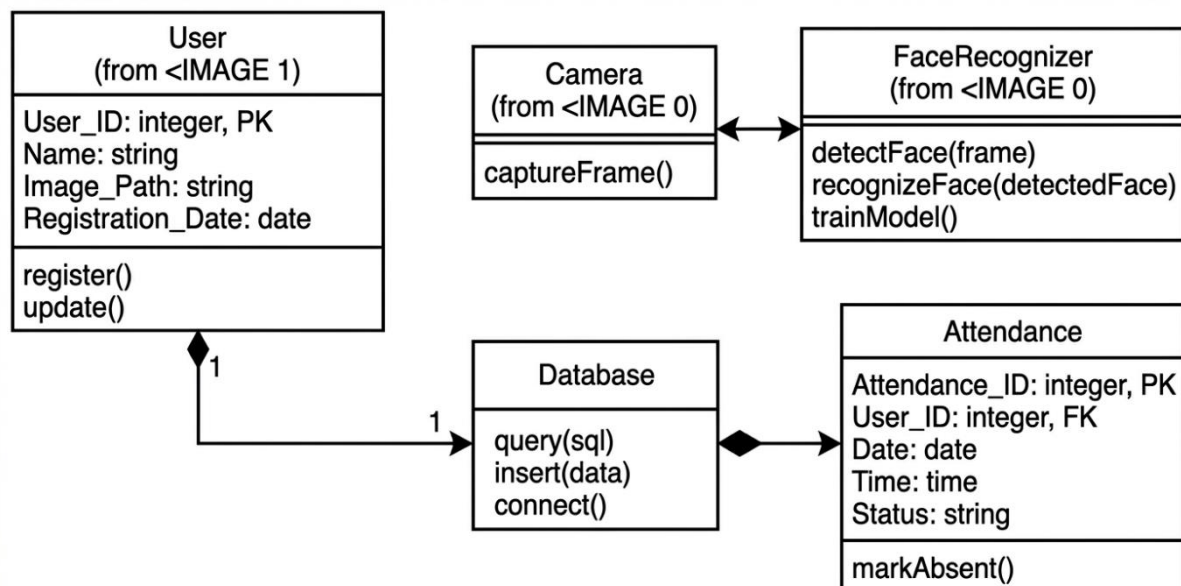
- View attendance
- Manage records

Class Diagram

Represents system classes and relationships.

Main Classes:

- User
- FaceRecognizer
- Attendance
- Database
- Camera



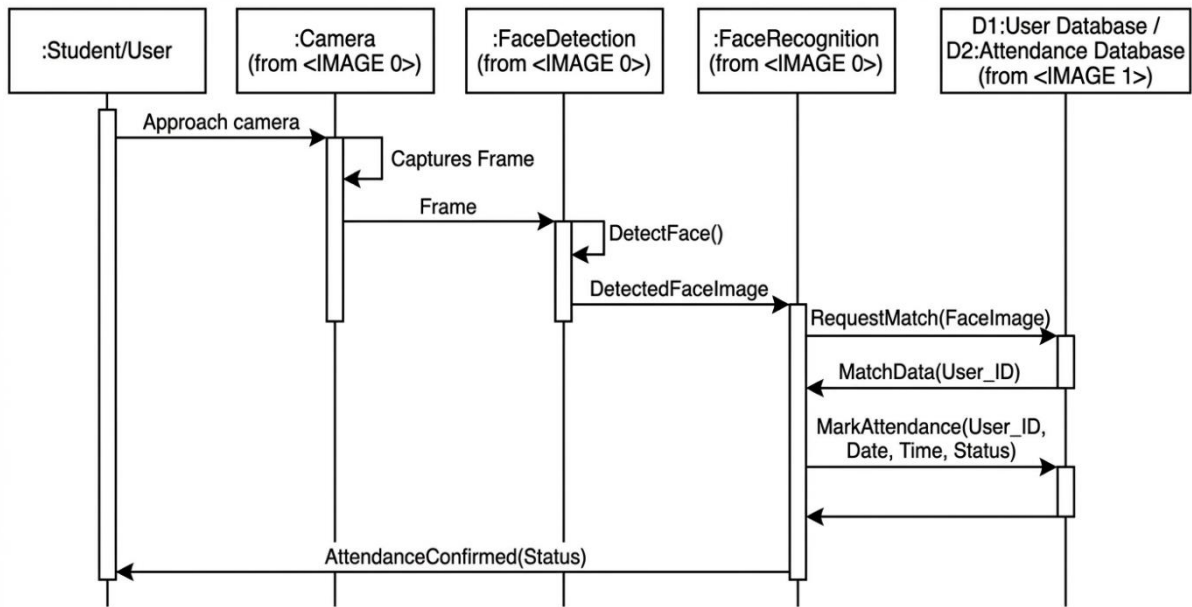
UML Class Diagram

Sequence Diagram

Shows sequence of steps in attendance process.

Steps:

User → Camera → Face Detection → Face Recognition → Database → Attendance Marked

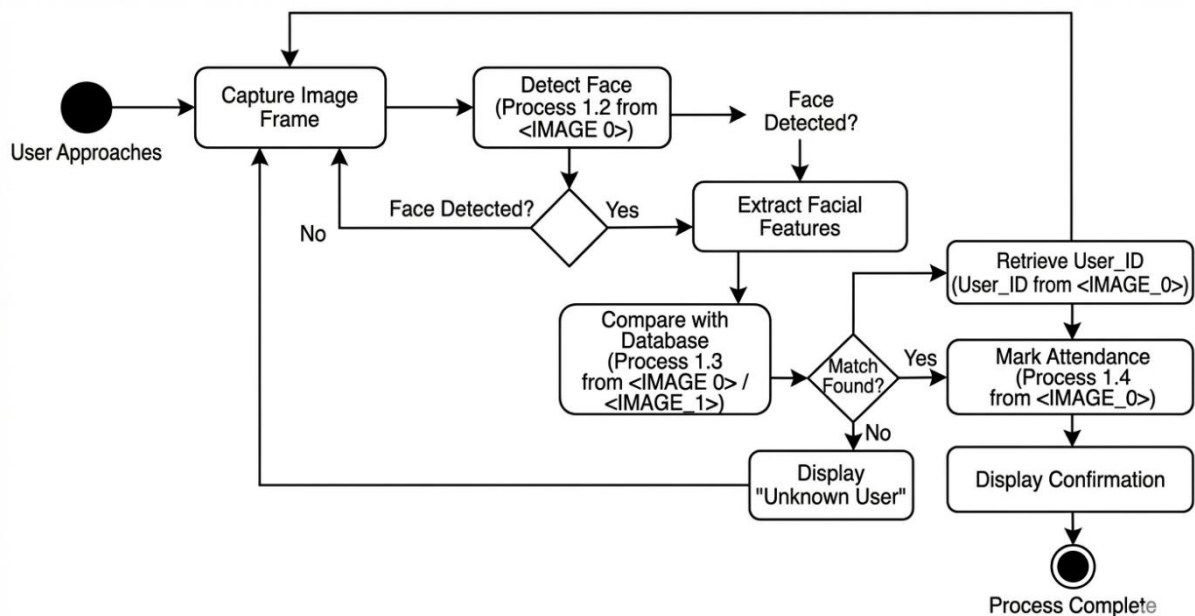


UML Sequence Diagram

Activity Diagram

Shows workflow of system.

Start → Capture image → Detect face → Recognize face → Mark attendance → End



UML Activity Diagram

Design Rationale:

UML diagrams are selected because they provide clear understanding of system behavior and structure. They help in planning system development and reduce confusion during implementation.

b. Database Design

The database stores user details and attendance records. A structured database helps in managing large amounts of data efficiently.

Database Type:

CSV file or SQLite database is used because it is simple and suitable for small to medium projects.

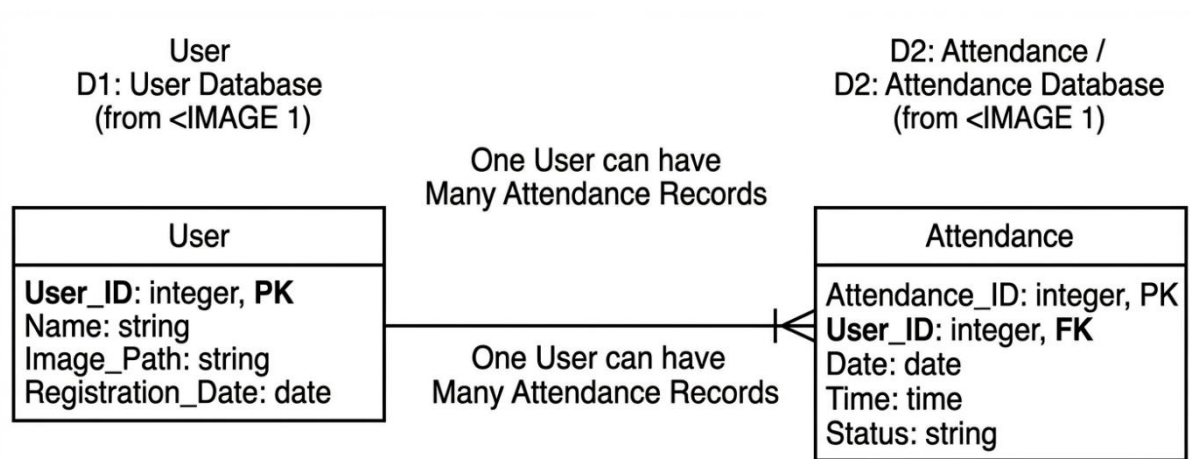
Main Tables:

1. User Table

Field Name	Description
User_ID	Unique ID
Name	Student name
Image_Path	Stored face image
Registration_Date	Date of registration

2. Attendance Table

Field Name	Description
Attendance_ID	Unique ID
User_ID	Student ID
Name	Student name
Date	Attendance date
Time	Attendance time
Status	Present/Absent



Entity Relationship ER Diagram

Database Design Rationale:

SQLite/CSV is selected because:

- Easy to implement
- No need for complex server setup
- Suitable for academic projects
- Fast data retrieval
- Simple data storage

Tools and Techniques Used:

Programming Language: Python

Libraries: OpenCV, NumPy, Face Recognition

Database: CSV / SQLite

IDE: VS Code / Jupyter Notebook

Operating System: Windows/Linux

These tools are selected because they are free, easy to use, and widely supported for AI-based projects.

Conclusion

The design of the AI Attendance System ensures efficient working of face recognition technology for attendance management. The architectural design, UML diagrams, and database structure provide a clear understanding of system components and data flow. The selected tools and techniques help in building a reliable, accurate, and

easy-to-use attendance system. The modular approach also allows future improvements such as cloud storage, mobile integration, and real-time monitoring.